RITHIKHA S

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Education

Vellore Institute of Technology, Chennai **Tamil Nadu**

Pursuing Bachelor of Technology in Artificial Intelligence and Robotics *Aug 2023 – Aug 2027*

Delhi Public School Bangalore East **Bangalore**

High Secondary – 78.8% *Jun 2011 – May 2023*

Relevant Coursework

- Artificial Intelligence
- Database Management
- Web Development
- Robot Programming
- Arduino
- Machine Learning

Honors and Industry Programs

Visteon Scholar – Visteon Scholar Program **Jan 2026 – Present**

- Selected for a competitive industry-led automotive software development program.
- Undergoing structured training in Embedded Systems, Automotive Electronics, and SDLC.
- Participating in technical assessments, hackathons, and mentorship sessions.

Projects

Manipulation of Robotic Arm | *ROS1, Python, Arduino IDE* **Mar 2025 – Apr 2025**

- Designed a ROS1-based publisher–subscriber architecture to control a multi-joint robotic arm via serial communication with Arduino.
- Implemented Python ROS nodes for joint command publishing and feedback subscription, enabling real-time angle manipulation and motion testing.
- Integrated control within a ROS workspace and validated communication latency and actuation responsiveness.

Quiz based To Do List Website for Students | ***HTML, CSS, JavaScript*** **Feb 2025 – May 2025**

- Developed a gamified task management platform where task completion unlocks quiz-based rewards to improve student engagement.
- Built responsive front-end architecture using modular JavaScript for task CRUD operations and progress tracking logic.

Braille to Text Transcription | *Arduino IDE* **Feb 2025 – May 2025**

- Designed a wearable haptic glove with six vibration motors to convert digital text into real-time Braille patterns for visually impaired users.
- Developed a button-based Braille input interface mapped to character encoding logic and displayed decoded output on an LCD module.

CURA – AI Powered Personal Journal | *Flask, Python, OpenAI* API **Aug 2025 – Nov 2025**

- Developed a full-stack wellness application enabling structured journal entry and emotion analysis using using polarity-based sentiment scoring models.
- Designed backend workflows in Flask for processing user inputs and generating reflective prompts.
- Experimented with voice input integration using speech-to-text APIs for future multimodal expansion.

Certifications

AWS APAC Solutions Architecture virtual experience program on Forage - September 2025

- Designed a simple and scalable hosting architecture based on Elastic Beanstalk for a client experiencing significant growth and slow response times.
- Described my proposed architecture in plain language ensuring my client understood how it works and how costs will be calculated for it.

Vista Equity Partners AI in Action Job Simulation on Forage - January 2026

- Completed a job simulation involving prompt engineering and GenAI workflow automation for Vista's Portfolio Operations team.
- Applied Vista's four-part prompt framework (Ask, Context, Examples, Desired Output) to synthesize customer NPS survey feedback into executive-ready insights, improving clarity and relevance of outputs by > 50% through iteration.
- Designed and tested repeatable GenAI workflows (news aggregation, data reconciliation, and qualitative insight

summarization) using tools such as ChatGPT and Copilot, ensuring accuracy, professional tone, and scalability.

- Evaluated AI-generated outputs for accuracy, tone, and usefulness; implemented guardrails and refinement techniques that reduced error rates and increased stakeholder usability of results.

Technical Skills

Languages: Java, Python, C, C++
AI/ML: Numpy, Pandas, OpenCV, YOLO
Robotics & Embedded: ROS1, Arduino, Linux
Web & Backend: HTML, CSS, JavaScript, React, Flask
Tools & IDE: Git, Google Colab, VS Code, Arduino IDE
Database: MySQL, DBMS Concepts

Extracurricular

Technocrats Robotics **Jun 2025 – Nov 2025**
Programming Member *VIT Chennai*

- Member of programming team of Technocrats Robotics for Robocon and Rover competitions.